Military Service and Social Tolerance: a DiD approach

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Abstract

Does military conscription shape deep-rooted social values? Leveraging the staggered abolition of mandatory military service across 21 European countries, we estimate its causal impact on tolerance towards homosexuality using European Social Survey (ESS) data. While a baseline Two-Way Fixed Effects (TWFE) model suggests a statistically significant effect, implying that draft exemption decreases tolerance, we show that this result is a methodological artifact. Applying the heterogeneity-robust estimator proposed by Callaway and Sant'Anna (2021), we find perfect pre-treatment parallel trends and a precisely estimated null effect post-reform. Our findings indicate that compulsory military service does not have a lasting causal impact on attitudes regarding sexual minorities, challenging the political narrative of the military as a "school of the nation" capable of creating social tolerance.

1 Introduction

In recent years, the reintroduction of mandatory military service has become a prominent topic in the European political debate. Driven by growing geopolitical tensions and perceived social fragmentation, policymakers across Europe are increasingly reconsidering conscription as a tool to forge civic virtues. Sweden, for instance, reinstated the draft in 2017. Simultaneously, right-wing populist parties, notably AFD in Germany, Rassemblement National in France and Lega in Italy, have actively campaigned for its reintroduction. These political movements frequently frame the military as a ”school of the nation”, assuming it acts as an essential disciplinary engine capable of restoring national identity, bridging social divides, and fostering social cohesion. This paper investigates whether mandatory military service truly has this causal impact on social tolerance, specifically focusing on attitudes towards homosexuality.

Theoretically, this mechanism of forced integration is deeply rooted in Allport's (1954) Contact Hypothesis, which posits that intergroup contact under conditions of equal status and shared goals effectively reduces prejudice. The military environment provides a unique and extreme application of this theory. By stripping recruits of their civilian socio-economic markers through uniform dress, standardising living conditions, and enforcing shared hardships, conscription artificially manufactures equal status. Young men are forced into prolonged cohabitation and must cooperate with peers from vastly different cultural and geographical backgrounds to achieve collective objectives. Recent empirical evidence supports this view in the context of conscription, demonstrating that the military draft shapes views towards immigrants by altering exposure to diversity (De Luca, Montalbano, and Stillman, 2025) and significantly impacts institutional trust (Bove, Di Leo, and Giani, 2022). Crucially, interpersonal contact is also highly effective in alleviating prejudice towards sexual minorities. A robust body of social psychology literature demonstrates that direct contact with homosexual individuals is one of the strongest predictors of favourable attitudes toward them, frequently surpassing the explanatory power of standard demographic variables such as education, religion, or political ideology (Herek and Glunt, 1993; Herek and Capitanio, 1996).

While the literature suggests that the draft can alter attitudes toward certain out-groups, its causal effect on deeper moral attitudes towards sexual minorities remains unexplored. We try to fill this gap by testing whether the forced integration of the military draft actually creates broader social tolerance, and consequently, whether abolishing the draft makes society less tolerant.

To answer this research question, we exploit the quasi-random variation generated by the staggered abolition of military conscription across 21 European countries. Using individual-level data from the European Social Survey (ESS), we employ a cohort-based Difference-in-Differences approach. To establish a robust counterfactual, we leverage women from the exact same birth cohorts, who were historically exempt from compulsory service across our sampled countries, as a clean, never-treated control group.

2 Data

For our empirical analysis, we employ data from all eleven the the European Social Survey (ESS) waves, a repeated cross-sectional survey. The ESS provides comprehensive individual-level data across several European countries, covering multiple social topics, including institutional trust, political preferences, human values, and detailed socio-demographic characteristics. To capture our main effect of interest, we focus on the key outcome variable named *freehms*[1]. This variable asks respondents to evaluate the statement "Gays and lesbians should be free to live life as they wish," with responses recorded on a 5-point Likert scale ranging from 1 ("Agree strongly") to 5 ("Disagree strongly").

The treatment assignment is determined by the specific year of birth. Following the timeline provided by Bove et al. (2022), we define the pivotal cohort for each country as the first generation of men exempted from mandatory military service due to national reforms. The key independent variable is a dummy indicator taking the value of 1 for cohorts born in or after the pivotal year, and 0 otherwise. Our geographical scope encompasses European countries, both treated (see Table 1) and untreated, including Austria, Cyprus, Denmark, Estonia, Finland, Greece, Norway, Switzerland, the Russian Federation, and Turkey. To ensure a valid counterfactual, we intentionally exclude countries that never implemented mandatory conscription, as these would mechanically fall into the control group despite lacking any comparable military tradition.

Finally, to mitigate both omitted variable bias and bad control issues, we carefully chose our set of covariates including only pre-determined variables that describe the respondent's environment at age 14, prior to any potential military draft. These controls include the father's and mother's educational levels and occupational status, as well as parental country of birth and minority ethnic group affiliation.

Table 1: Timeline of military service suspension

Country	Year of reform	Pivotal Cohort	Country	Year of reform	Pivotal Cohort
Albania	2010	1991	North Macedonia	2006	1988
Belgium	1995	1976	Netherlands	1997	1979
Bulgaria	2008	1989	Poland	2009	1990
Croatia	2008	1990	Portugal	2005	1986
Czech Republic	2005	1986	Romania	2006	1988
France	2002	1983	Slovak Republic	2006	1987
Germany	2012	1994	Slovenia	2004	1985
Hungary	2005	1986	Spain	2002	1983
Italy	2005	1986	Sweden	2011	1992
Latvia	2006	1988	United Kingdom	1961	1942
Luxembourg	1967	1948			

Source: Bove et al., 2022

[1]While the ESS provides several measures of social attitudes, we focus on *freehms* as it represents a core indicator of tolerance towards marginalized groups.

3 Empirical Strategy

We proceed by steps, initially implementing a baseline specification as a benchmark. Then, we unpack the average effect under the more credible hypothesis of dynamic Average Treatment Effects. Finally, we compare these results with the most robust approach: the Callaway and Sant'Anna (2021) estimator, which allows for the precise estimation of heterogeneous treatment effects in a context of staggered adoption, as in our scenario.

3.1 Identification Strategy: A Cohort-Based Approach

Our identification strategy relies on a cohort-based Difference-in-Differences (DiD) approach. Unlike a standard DiD where the temporal running variable is the calendar year of the observation, our timeline is defined by the respondent's birth cohort. The "treatment" in our context is the exemption from mandatory military service. Because national governments abolished or suspended the draft in different years, the exposure to this treatment is staggered across both countries and birth cohorts.

A fundamental challenge in evaluating national macroeconomic or social reforms is finding a valid counterfactual within the same country. To isolate the causal effect of draft abolition, we exploit a unique historical institutional feature: women were universally exempt from mandatory military service across all countries in our sample. Consequently, we utilize women from the exact same birth cohorts as a clean, never-treated control group.

The core identifying assumption for our empirical strategy is the parallel trends assumption across cohorts. Specifically, we assume that, in the absence of the military reforms (i.e., the draft had remained mandatory for all men), the baseline gap in tolerance towards homosexuality between men and women would have evolved in parallel across successive generations. By plotting dynamic event-study estimates, we can visually test and confirm the validity of this assumption for the pre-treatment cohorts.

3.2 The Baseline TWFE Model

The baseline specification relies on the standard Two-Way Fixed Effects (TWFE) estimator. We estimate the following equation:

$$Y_{icwt} = \alpha + \beta(Male_i \times Post_{ct}) + \delta Male_i + \gamma X_i + \mu_c + \lambda_t + \phi_w + \varepsilon_{icwt}$$

where Y_{icwt} is the tolerance level towards homosexuality of individual i, born in year t, in country c, and interviewed during survey wave w. $Male_i$ is a dummy equal to 1 for men, and $Post_{ct}$ is a dummy equal to 1 if the cohort t in country c was exempted from the draft. The coefficient of interest is β, which captures the Average Treatment Effect on the Treated (ATT) under the strict assumption of homogeneous treatment effects.

Crucially, the β parameter allows us to empirically test the Contact Theory mechanism. If the military truly acts as an integrating "school of the nation," removing this forced social contact should

reduce tolerance. Because our outcome variable is measured on a reversed Likert scale (where higher values indicate lower tolerance), the Contact Theory implies that β should be positive.

We include a vector of pre-determined socio-economic controls at age 14 (X_i), country fixed effects (μ_c), birth-year fixed effects (λ_t), and survey wave fixed effects (ϕ_w). The inclusion of wave fixed effects is crucial to absorb general, Europe-wide cultural shifts in tolerance occurring at the time of the interview. Standard errors are clustered at the country level to account for serial correlation within countries.

3.3 Event Study Analysis

To assess the validity of the parallel trends assumption and explore the dynamic effects of the reform, we expand our baseline model into an event-study framework:

$$Y_{icwt} = \alpha + \sum_{\tau=-10,\tau\neq-1}^{10} \beta_\tau (Male_i \times \mathbb{I}\{t - E_c = \tau\}) + \delta Male_i + \gamma X_i + \mu_c + \lambda_t + \phi_w + \varepsilon_{icwt}$$

where E_c is the pivotal cohort year in country c, and τ denotes the relative distance in years from the reform. We omit the period $\tau = -1$, meaning all coefficients β_τ are estimated relative to the cohort born exactly one year before the draft abolition. Consistent with the baseline model, we maintain the full set of fixed effects (μ_c, λ_t, ϕ_w) and socio-economic controls (X_i).

3.4 Callaway and Sant'Anna (2021) Estimator

Recent econometric literature highlights severe limitations in the standard TWFE estimator when treatment adoption is staggered across groups and treatment effects are heterogeneous over time (De Chaisemartin and D'Haultfœuille, 2020; Goodman-Bacon, 2021). TWFE estimates can be severely biased due to negative weighting driven by forbidden comparisons between newly treated and already-treated units. To address this critical bias, we implement the heterogeneity-robust estimator developed by Callaway and Sant'Anna (2021). This method avoids forbidden comparisons by strictly estimating group-time average treatment effects, $ATT(g,t)$, using as a control group the never-treated units, specifically women, who were historically exempt from conscription and, where applicable, individuals from not-yet-treated cohorts. We then aggregate these $ATT(g,t)$ parameters into an event-study plot to evaluate the true dynamic causal effect of draft abolition without the contamination of negative weights.

4 Results

4.1 Static TWFE Estimates

As shown in Table 2, the baseline model points to a positive and statistically significant result at the 5% level. A positive coefficient implies that the abolition of mandatory military service for men would, on average, decrease their tolerance towards homosexuality. Still, the economic magnitude of this

effect is only marginal: a coefficient of 0.059 translates to a variation of roughly 1.5 percent of the entire 4-point scale range. Importantly, these estimates remain robust to the progressive inclusion of the pre-determined socio-economic controls at age 14. Furthermore, we recognize that this TWFE coefficient represents only a static average across all periods and cohorts, prompting the need for a dynamic analysis.

Table 2: Naive TWFE Effect of Draft Abolition on Tolerance

	(1) Baseline	(2) + Father Edu	(3) + Both Parents	(4) + All Controls
Male × *Post-Treatment*	0.059**	0.055**	0.052**	0.054*
	(0.024)	(0.025)	(0.025)	(0.031)
Male	0.172***	0.174***	0.174***	0.204***
	(0.015)	(0.021)	(0.021)	(0.024)
Post-Treatment	0.069	0.088**	0.087**	0.032
	(0.042)	(0.037)	(0.037)	(0.041)
Observations	438,850	149,670	146,558	65,530
Within R^2	0.008	0.013	0.015	0.021

Notes: Standard errors in parentheses, clustered at the country level. All specifications include country, birth-year, and survey wave fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.2 Dynamic Effects and Pre-Trend Testing

To investigate the dynamic effects and test the parallel trends assumption, we plot the event-study coefficients in Figure 1, with 95% confidence bands. As illustrated in the graph, the pre-treatment estimates are closely clustered around zero and lack statistical significance, confirming the absence of pre-existing trends in tolerance between our treatment group (males) and our control group (females). Furthermore, the post-treatment coefficients reveal no statistically significant effect, indicating that the abolition of the military draft had no impact on social values such as tolerance. This dynamic evidence suggests that our baseline positive TWFE result was likely a statistical artifact driven by the estimator's limitations in staggered adoption settings.

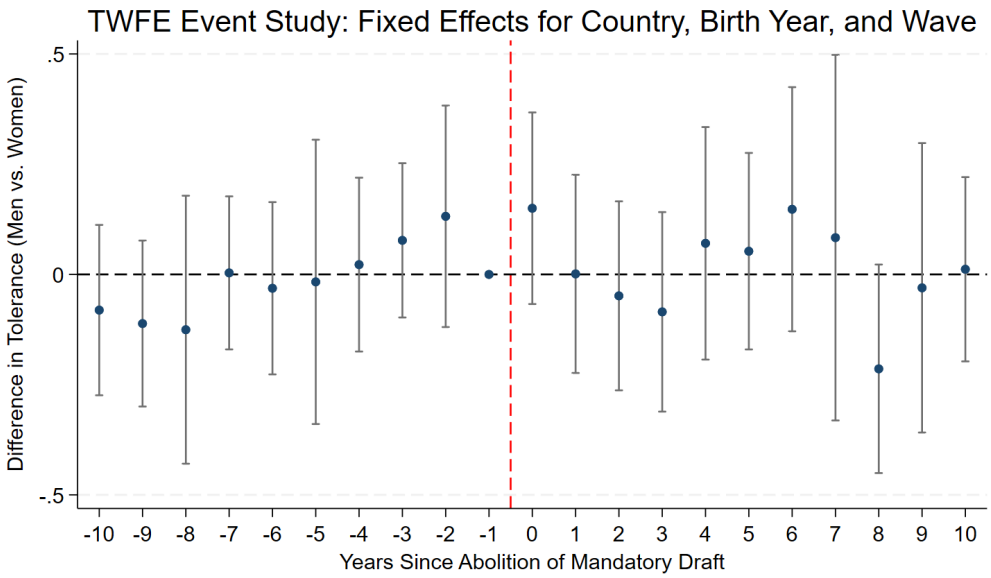

Figure 1: Event Study Estimates of Draft Abolition on Tolerance

4.3 Robustness Check: The Callaway and Sant'Anna Approach

These results support our suspicion that the baseline findings were entirely driven by methodological biases. As established in the recent econometric literature, standard TWFE models in staggered adoption settings mechanically perform forbidden comparisons between newly treated and already-treated units, thereby providing biased estimates of the treatment effect. By overcoming this structural limitation, the Callaway and Sant'Anna (2021) approach confirms the pattern suggested by our initial event study: once dynamic and heterogeneous treatment effects are properly accounted for, there is no empirical evidence that the abolition of the military draft had any causal impact on social tolerance towards homosexuality.

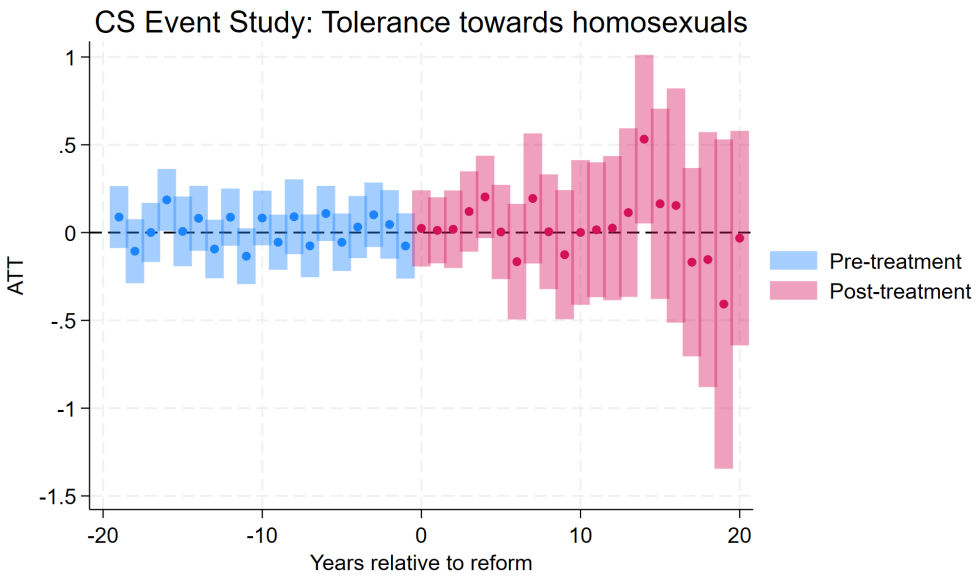

Figure 2: Dynamic Effects of Military Service Abolition on Social Tolerance

5 Conclusion

This project investigated the political and sociological premise that mandatory military service acts as an engine for social integration, specifically testing its causal impact on tolerance towards homosexuality. Initially, our naive Two-Way Fixed Effects estimates appeared to validate Allport's Contact Theory, suggesting that abolishing the draft significantly decreased tolerance.

However, recognizing the severe biases of TWFE models in staggered adoption settings, we applied the heterogeneity-robust estimator proposed by Callaway and Sant'Anna (2021). Once the forbidden comparison issues are properly corrected, the apparent effect completely disappears. We find perfect pre-treatment parallel trends and a precisely estimated null effect. The end of compulsory military service did not lead to any significant shift in social attitudes towards sexual minorities across Europe.

These findings deliver a crucial lesson for current European political debates: we find no empirical evidence to support the narrative that the military serves as a "school of the nation" capable of engineering deep-rooted moral values. While forced military integration may impact other dimensions of civic life, its relatively short-lived nature is insufficient to alter fundamental social attitudes regarding sexual minorities. Ultimately, our results suggest that such deep-seated values are formed through broader cultural and educational channels, entirely independent of military conscription.

References

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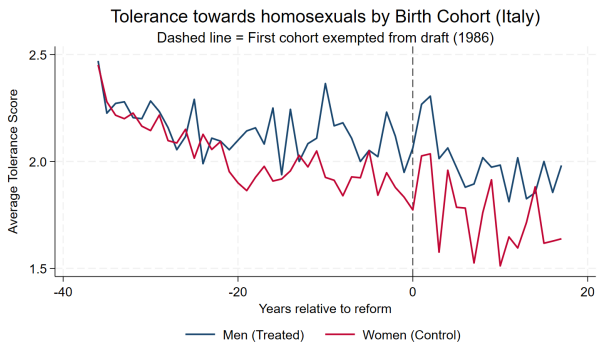
[9] Herek, G. M., & Capitanio, J. P. (1996). "Some of my best friends": Intergroup contact, concealable stigma, and heterosexuals' attitudes toward gay men and lesbians. *Personality and Social Psychology Bulletin*, 22(4), 412-424.

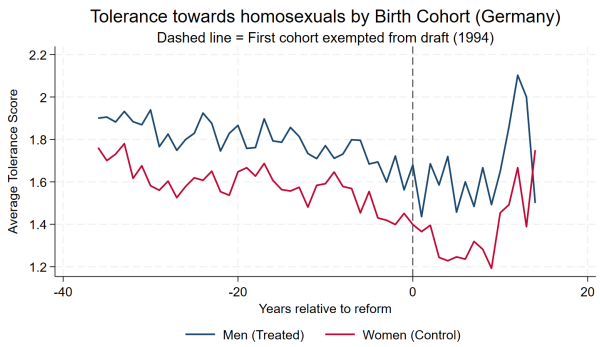
A Appendix

A.1 Descriptive Evidence: Heterogeneity across Countries

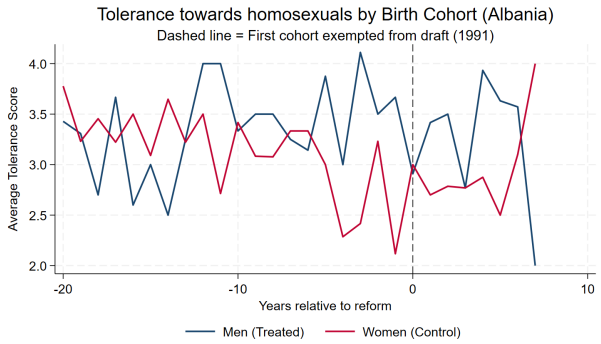
To illustrate the underlying data structure and the variation used in our analysis, the following graphs present the average tolerance scores by birth cohort for a selection of representative countries.



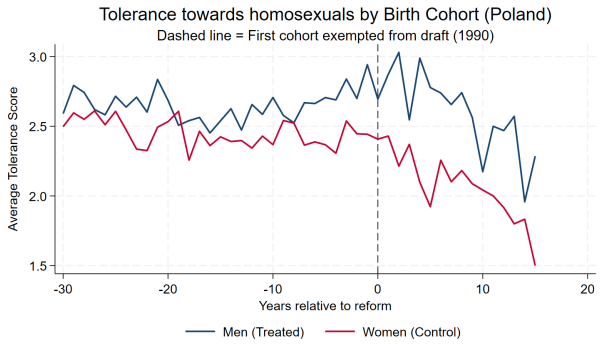
(a) Italy (1986)



(b) Germany (1994)



(c) Albania (1991)



(d) Poland (1990)

B Software and Replication

All econometric estimations were performed using Stata 18 (StataCorp, 2023). To ensure full transparency and replicability, the full code and datasets used in this project are hosted on the following GitHub repository:

https://github.com/lorenzodelcol99/Impact-Evaluation-Project